ICT374 Operating Systems and Systems Programming

Review and Examination

30th October 2025

Review of the Unit

This unit is about operating systems – the most important piece of software in any computer system. We have approached the subject from two perspectives – OS design and theory and OS programming.

In OS design and theory part (topics 1, 2, 3, 9, 10, and 11), we have looked at the OS from its inside. We have discussed

- what the aims of an operating system are;
- how such aims are achieved through combinations of hardware and software;
- important components of an operating system kernel: process management, file systems, memory management, and processor scheduling;
- various conceptual models, data structures and algorithms used in designing these components, and their strength and weakness;
- concurrency related issues such as race conditions, mutual exclusion, deadlocks, starvations, multi-threading, symmetric multi-processing, microkernel, and so on.

We have used real world operating systems such as UNIX (including Linux) and Microsoft Windows as examples in our discussions. Specifically, we have covered the following important topics:

- Hardware support for operating systems;
- History of operating systems;
- Process management;
- File system management;
- Memory management;
- Processor scheduling;
- Concurrency related issues.

It is hoped that these topics will provide you with a solid basis for understanding how an operating system works internally. They should also give you a foundation for further in-depth study of operating systems and for learning and adapting to future technological changes in the area of operating systems.

In the second part of this unit (topics 4, 5, 6, 7, and 8), we have looked at the OS from its outside. We have leant to appreciate how a set of complex hardware components is turned into a powerful abstract machine that is simple, elegant and easy to use. We have learnt to create application programs on this abstract machine. We have focused on the POSIX programming interface (which are supported by all UNIX-like operating systems, including Linux, Mac OS X

Darwin). We have learnt how to control processes, how to manipulate files, how to handle signals, how to communicate with other processes and how to write network programs using sockets. The topics covered are:

- Process programming
- File system programming
- Signal handling
- Interprocess communication
- Network programming

The second part is very practical. It opens up a new world of possibilities that were not available until now. Of course, the number of system calls and functionalities we have covered in this unit is only a small (but very important) part of UNIX programming interface. This should lay a solid foundation for UNIX programming.

Through 9 weekly labs, two major assignments, one of which is a very substantial programming project, we hope that you have now gained some real-world programming experience using C (or C++). To many who are not very experienced in programming, we hope this is a hard but rewarding experience.

The Final Examination

The final exam will be a 2-hour closed book exam, consisting of multiple-choice questions, small programming questions and theory/conceptual questions, similar to the sample exam paper posted in the LMS.

Date: Thursday, 13 November 2025
Time: 9:00 am
Venue: 415 (Gym)

Please check the University's Exam Timetable to find the time and venue of the exam and make sure that the above details are correct and there is no last-minute change to the exam time or venue.

The final examination covers both OS design and theory, and Unix systems programming, worth approximately 50% each. There are two parts in the examination.

All 11 topics could be examined. A copy of the front page of the exam paper is given at the end of this document.

Part I consists of 20 multi-choice questions. Each question is worth 1 mark, with a total of 20 marks. This part covers the OS theory and UNIX programming concepts. Some questions in Part I are related to Unix system call interface. In each question, you are given 4 choices labelled between a to d. You must mark your answer to each question in the Computer Answer Sheet (not on the Question Book). You must use a blue or black pen, not a pencil, not a red pen, to mark your choice. You must select your choice that either ***best answers*** the question or ***best completes*** the question. Please watch the following video to familiarize yourself with the way the multi-choice questions must be answered:

- Computer Answer Sheet Instructions

Part II consists of 6 questions, focusing on Unix systems programming. This part is worth 80 marks.

The first three questions in Part II (40 marks) ask you to write either a complete program or a program snippet using C and/or UNIX system calls. There could be other forms of questions, such as questions that ask you to provide a general solution to a problem using a pseudo-code. Many of these questions require you to use some of the most important UNIX system calls. It is therefore important that you understand and remember the syntax and semantics of the most fundamental system calls, such as `fork`, `exec`s, `wait`, `exit`, `read`, `write`, `open`, `close`, `socket`, `connect`, `accept`, `listen` etc.

In case you cannot recall the syntax of a system call/function, you may use your own notation for the system call/function and provide a note for the system

call/function, indicating its purpose and the required parameters. This will help us to judge how well you are able to solve the problem if you were given access to the relevant reference documentation and to decide the mark that is fair and adequate for the question.

Questions 4 to 6 focus on operating system theory and they are worth of 40 marks.

Altogether, the Question Paper consists of 26 questions with a total of 100 marks. You should attempt all questions.

For questions in Part II, you must write your answer directly in the space below each question in the Question Paper. There is no separate answer book. If the space under each question is not enough, you can continue your answer in one of three blank pages at the end of the Question Paper.

The examination is closed-book examination, so books, calculators, mobile phones etc are not allowed. However, you need to bring at least one black pen or one blue pen (better still, bring several pens with you in case one pen fails during the exam).

To prepare for the exam, I recommend that you thoroughly read the relevant chapters of textbook and the lecture notes for each topic and review your solutions to both the C and UNIX programming exercises as well as the theory exercises for each weekly lab.

In addition to read the lecture notes and textbooks, it is recommended that you also try the sample exam paper available from the Unit LMS.

ICT374 Operating Systems and Systems Programming
Final Examination

Semester 2, 2025

Time Allowed 2 hours plus 10 minutes reading time

INSTRUCTIONS

1. There are 12 pages in this Paper, including 3 blank pages at the end.
2. The questions are divided into two parts with 100 marks.
3. Part I consists of 20 multiple-choice questions, worth 1 mark each and 20 marks in total.
4. Part II consists of 6 questions worth a total of 80 marks.
5. Attempt all questions.
6. For questions in Part I, mark your answer to each question on the separate Computer Answer Sheet, not on this Question Book. You must use a blue or black pen (not a pencil) to mark.
7. For questions in Part II, write your answer directly in the space under each question.
8. If more space is required, continue your answer on the blank pages at the end. Make clear on the original page that your answer continues on a blank page, and give the page number.
9. Use the back of each page for rough notes if required, but do not write your answer on any back page, as anything written on it will be ignored.

Your Name: _____Student Number_____

Your Signature: _____

EXAMINATION AID ALLOWED

Provided by the University

Nil

Provided by the Candidate

Closed Book Examination – No Calculators Permitted